

Ahmed GHALEB

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 Chapter 4 System Design (System Development Based) / Experimental Setup (Research Based)

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CHAPTER 4

REQUIREMENT ANALYSIS AND DESIGN



4.1 Introduction

This chapter delineates the functional and non-functional requirements for the Al-Muneer Online Portal. It also elaborates on the system architecture, database schema, user interface design principles, and essential process flows. The main objective of this phase, carried out within PSM1, is to convert the stakeholder requirements and project goals identified in previous phases into a detailed blueprint for the system's development in PSM2. The design focuses on delivering an intuitive and effective experience for both potential clients (Visitors) seeking information about Al-Muneer Hall and the administrator (Owner) overseeing the portal's content and operations. The requirements and design components recorded in this document will serve as the foundation for the Software Requirements Specification (SRS) and System Design Document (SDD).

4.2 Requirement Analysis

Requirement analysis involves delineating the users' expectations and the system's characteristics. In the case of the Al-Muneer Online Portal, the requirements were primarily collected through direct engagement with the stakeholder (Mr. Almunajid, the proprietor of Al-Muneer Hall), as elaborated in the Case Study (Chapter 2.2.4). Additionally, an examination of standard functionalities anticipated from comparable online venue management systems was conducted, specifically adapted to the local context of Yemen.

4.2.1 Functional Requirements

Functional requirements describe what the system should do. For the Al-Muneer Online Portal, these are categorized by user roles and key system operations:

FR1: Visitor Functionalities

- **FR1.1 View Venue Information:** The system shall display comprehensive information about Al-Muneer Hall, including descriptions, capacity, services offered, and location details.
- **FR1.2 View Media Gallery:** The system shall allow visitors to browse a gallery of images and potentially videos showcasing the hall.
- **FR1.3 Check Availability:** The system shall provide an interactive calendar allowing visitors to view dates that are already booked and those that are available.
- **FR1.4 View Pricing Information:** The system shall display pricing information. This may include base rates and options for customizable packages or services that affect the price.
- **FR1.5 View FAQ:** The system shall display a list of frequently asked questions and their answers.
- **FR1.6 Submit Booking Inquiry:** The system shall provide a form for visitors to submit a booking inquiry, including their name, contact details, desired date(s), and any specific requests.
- **FR1.7 Submit Payment Proof (Optional):** The system shall allow visitors, after submitting an inquiry or upon instruction from the admin, to optionally upload an image file (e.g., screenshot of a local transfer receipt) as proof of payment. The system shall only accept common image formats (e.g., JPG, PNG) and enforce a reasonable file size limit.
- **FR1.8 Submit Feedback:** The system shall provide a form for visitors to submit feedback about the venue or their experience with the portal.

FR2: Administrator Functionalities

- **FR2.1 Secure Admin Login:** The system shall require administrators to log in using unique, authenticated credentials (username and hashed password). The system shall display an error message if authentication fails.
- **FR2.2 Manage Venue Information:** The system shall allow the administrator to create, update, and delete textual information about the hall (descriptions, services, FAQs).
- **FR2.3 Manage Media Gallery:** The system shall allow the administrator to upload new images/videos to the gallery and delete existing ones.
- **FR2.4 Manage Availability Calendar:** The system shall allow the administrator to mark dates as booked or available on the interactive calendar.
- **FR2.5 Manage Pricing:** The system shall allow the administrator to set and update base pricing and any customizable pricing options.
- **FR2.6 Manage Booking Inquiries:** The system shall allow the administrator to view submitted booking inquiries.
- **FR2.7 Manage Payment Proofs:** The system shall allow the administrator to view uploaded payment proofs and update the status of a booking (e.g., "Payment Verified," "Pending Confirmation").
- **FR2.8 Manage Feedback:** The system shall allow the administrator to view submitted user feedback.
- **FR2.9 Generate Basic Reports:** The system shall allow the administrator to view basic reports, such as a list of inquiries within a date range, a summary of booking statuses, or an overview of feedback received.
- **FR2.10 Manage Notifications:** The system shall allow the administrator to manage pre-defined message templates for notifications (e.g., inquiry received, payment proof acknowledged) and initiate communication via preferred channels—specifically WhatsApp through deep-link triggers, with email as a secondary option.

FR3: General System Functionalities

- **FR3.1 Responsive Design:** The system shall be responsive and accessible on common web browsers across various devices (desktops, tablets, smartphones).

- **FR3.2 Secure Data Handling:** The system shall securely handle all user-submitted data, including personal information in inquiries, uploaded payment proofs, and feedback.
- **FR3.3 Notification Generation:** Based on admin configuration and specific system events (e.g., new inquiry, payment proof submission), the system shall generate notifications to be sent to relevant parties (client and/or admin).

4.2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

- **NFR1 Performance:** The system should load pages within an acceptable time frame (e.g., 3-5 seconds on a typical internet connection). The interactive calendar should respond promptly to user interactions.
- **NFR2 Usability:** The portal should be intuitive and easy to navigate for both visitors and the administrator. The language used will be primarily Arabic, with an option for English if deemed feasible within the project scope. UI elements should be clear and user-friendly, considering local cultural norms for design.
- **NFR3 Security:**
 - **NFR3.1 Admin Authentication:** The admin login shall be protected against common vulnerabilities. Passwords must be hashed.
 - **NFR3.2 Data Protection:** User-submitted data and uploaded files (payment proofs) must be protected from unauthorized access.
 - **NFR3.3 SSL/TLS:** All communication between the client and server shall be encrypted using HTTPS.
- **NFR4 Reliability:** The system should be available for use with minimal downtime. Database backups should be considered as part of operational reliability.

- **NFR5 Maintainability:** The code should be well-structured, commented, and follow the chosen MVC pattern to facilitate future updates or modifications (relevant for PSM2 and beyond).
- **NFR6 Scalability (Basic):** While not expecting massive concurrent users initially, the system architecture (Spring Boot, chosen database) should be capable of handling a moderate increase in traffic (up to 10,000 monthly visitors as per initial scope) without significant performance degradation.

4.2.3 Use Case Table and Diagram

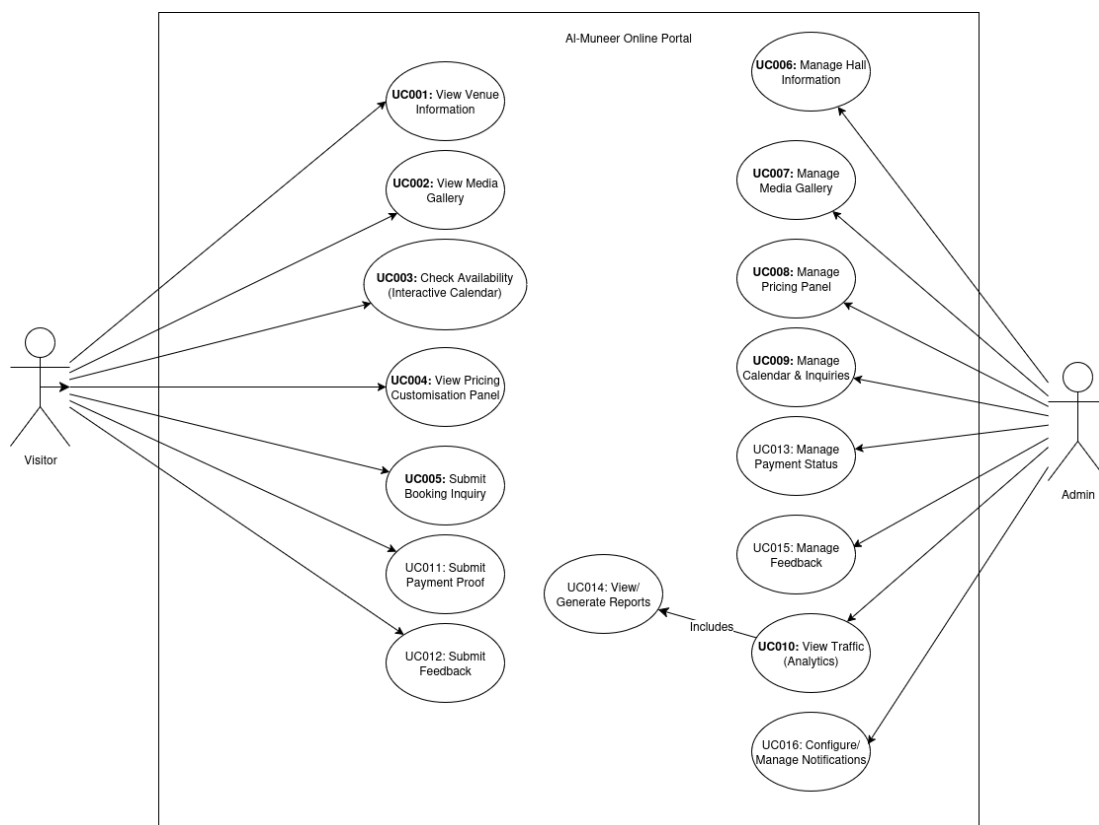


Figure 4.1: Use Case Diagram for the Al-Muneer Online Portal

4.2.3.1 Selected Use Case: Submit Booking Inquiry (UC005)

Table 4.1: Use Case Description for 'Submit Booking Inquiry' (UC005)

Attribute	Description
Use Case ID	UC005
Use Case Name	Submit Booking Inquiry
Actor(s)	Visitor
Pre-condition(s)	Visitor is on the Al-Muneer Online Portal and has navigated to the inquiry section/form.
Normal Flow	1. Visitor accesses the booking inquiry form. 2. Visitor fills in required fields (e.g., name, contact, desired date, event type, number of guests, specific requests). 3. Visitor submits the form. 4. System validates the input data. 5. System saves the inquiry details to the database. 6. System displays a success message to the Visitor. 7. System triggers a notification to the Administrator about the new inquiry.
Alternative Flow	4a. Invalid Input Data: 4a1. System displays an error message indicating the fields with invalid data. 4a2. Visitor corrects the data and re-submits (back to step 3). 7a. Notification Failure: 7a1. System logs the notification failure but still saves the inquiry and shows success to the visitor. Admin will see the inquiry upon login.
Post-condition(s)	• Booking inquiry is successfully recorded in the system. • Administrator is notified of the new inquiry.

- Visitor receives confirmation that their inquiry has been submitted.

4.2.3.2 Sequence Diagram (Submit Booking Inquiry - UC005)

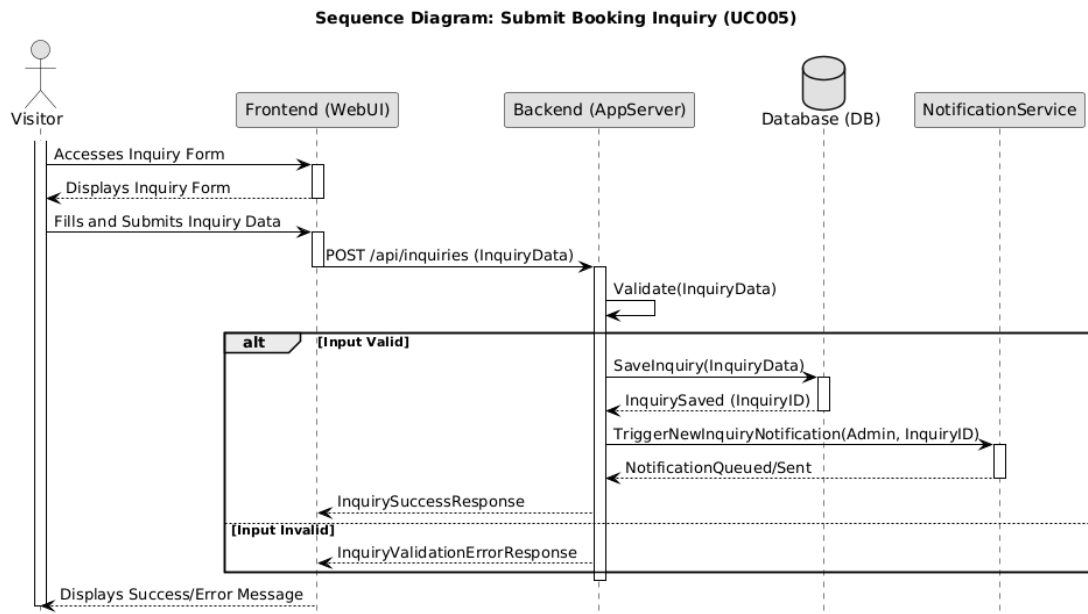


Figure 4.2: Sequence Diagram for 'Submit Booking Inquiry' (UC005)

4.2.3.3 Activity Diagram (Submit Booking Inquiry - UC005)

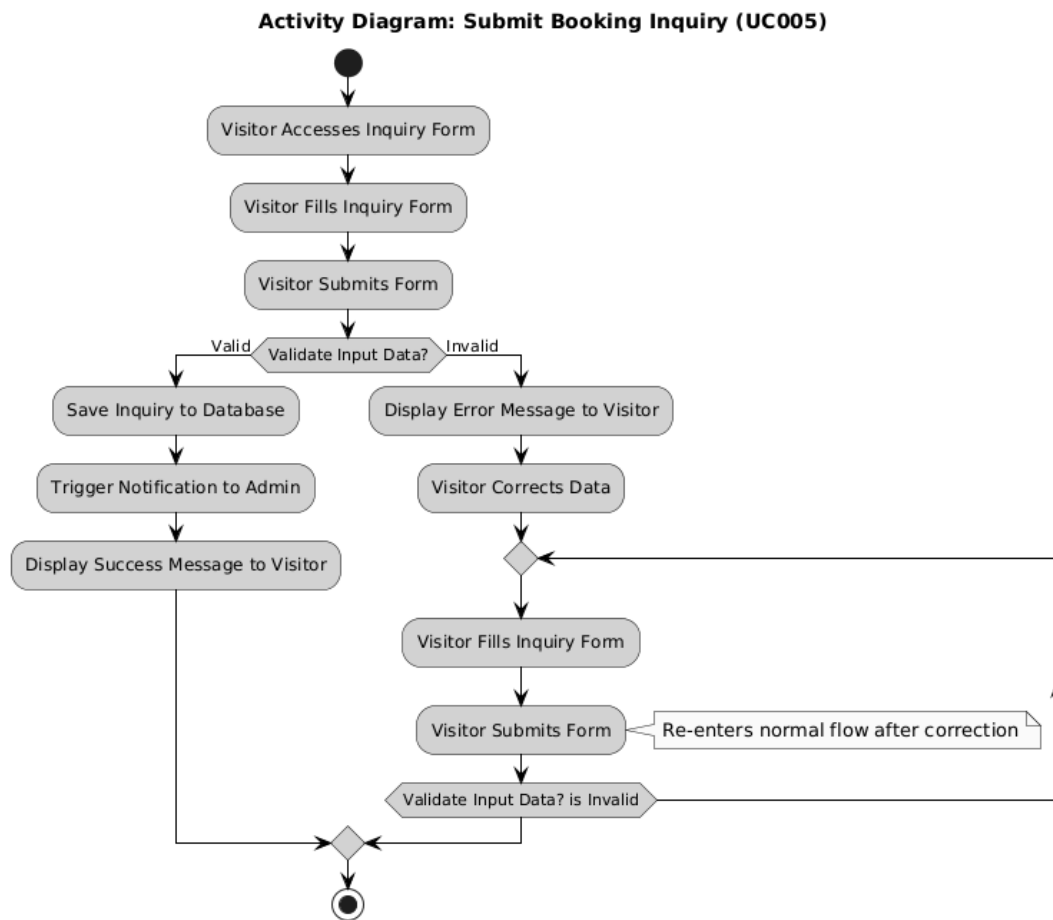


Figure 4.3: Activity Diagram for 'Submit Booking Inquiry' (UC005)

4.3 Project Design

The Al-Muneer Online Portal utilizes a Model-View-Controller (MVC) architectural framework within a singular, monolithic application. This structure guarantees a clear delineation of responsibilities: the Model (which encompasses data and business logic), the View (representing the user interface), and the Controller (which manages user input and the flow of the application). This cohesive strategy is exceptionally well-suited to the project's needs, facilitating both development and deployment.

4.3.1 System Architecture Diagram

The diagram presented below depicts the overarching Model-View-Controller (MVC) architecture utilized by the AI-Muneer Online Portal. Within this unified framework, the Spring Boot application is responsible for managing both the business logic and the rendering of the user interface, thereby establishing a coherent and sustainable system.

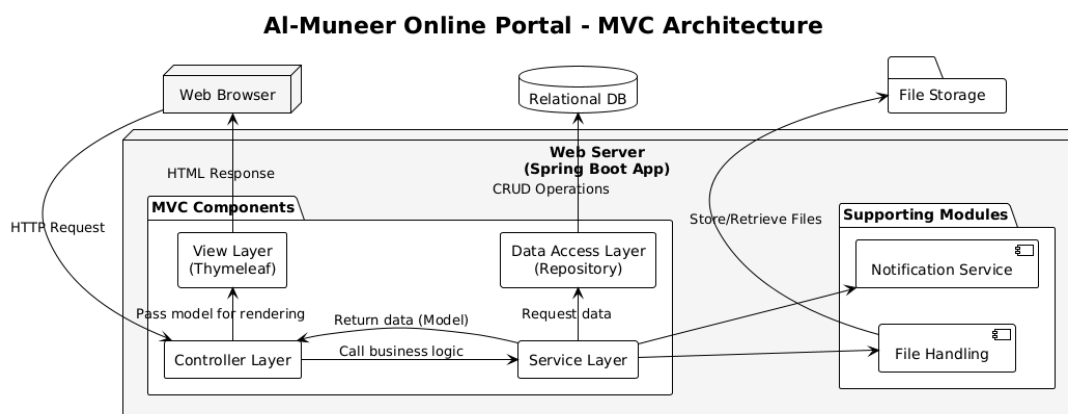


Figure 4.4: System Architecture Diagram (MVC)

4.3.2 Database Design

A relational database model is selected because of the organized characteristics of the data. The subsequent illustration is a basic conceptual Entity-Relationship Diagram (ERD). Primary Keys (PK) and Foreign Keys (FK) will be specified in a comprehensive design.

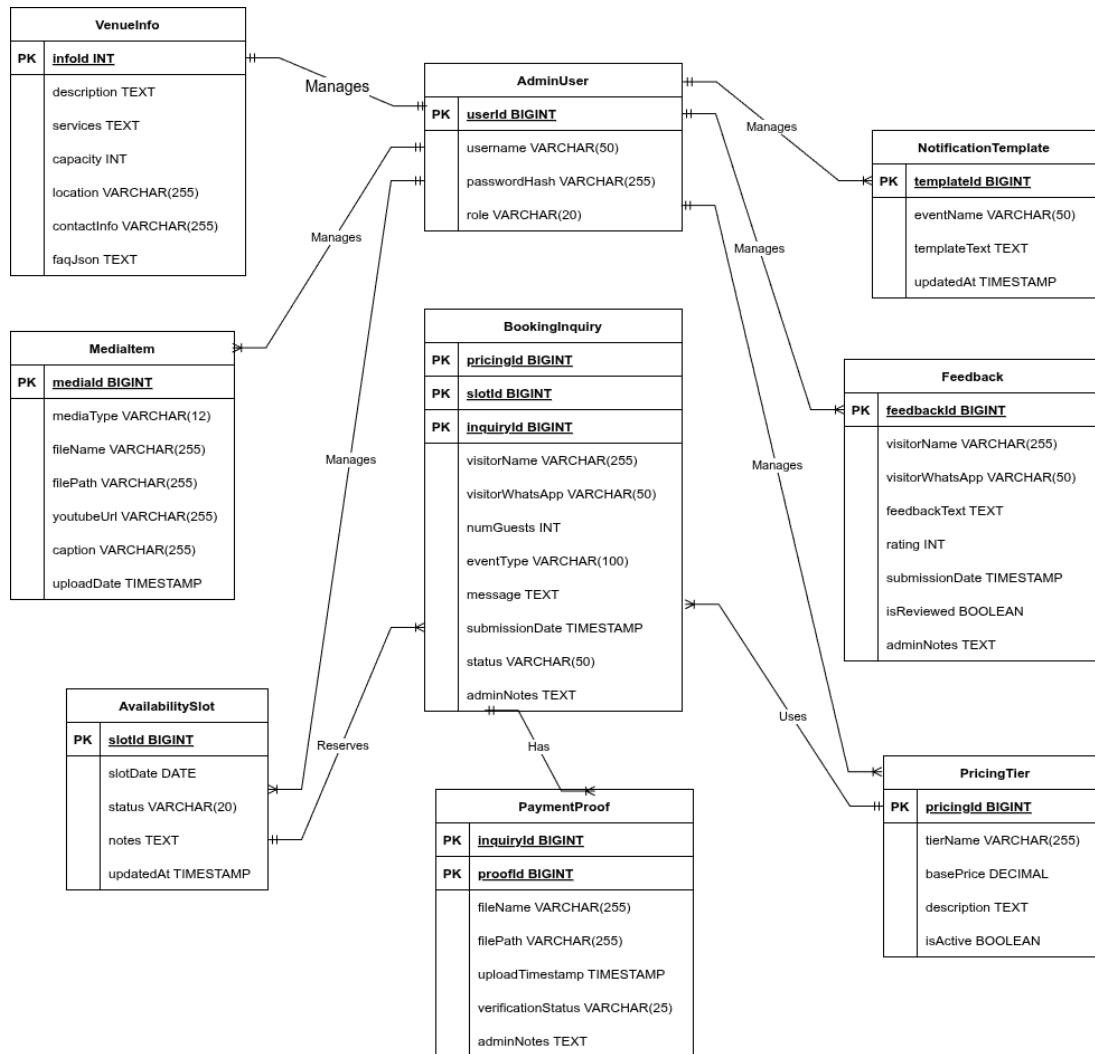


Figure 4.5: Conceptual Entity-Relationship Diagram (ERD)

Data Components:

- **AdminUser:** Stores admin credentials and roles.
- **VenueInfo:** General descriptive information about the hall.
- **MediaItem:** Details about uploaded images and videos for the gallery.
- **AvailabilitySlot:** Represents specific dates and their booking status.
- **PricingTier:** Information about different pricing packages or rates.
- **BookingInquiry:** Details of inquiries submitted by visitors.

- **PaymentProof:** Metadata for uploaded payment proof files, linked to inquiries.
- **Feedback:** Stores feedback submitted by users.
- **NotificationTemplate:** Stores pre-filled message templates for admin-triggered WhatsApp notifications.

4.4 Interface Design (Low-Fidelity)

4.4.1 Home Page

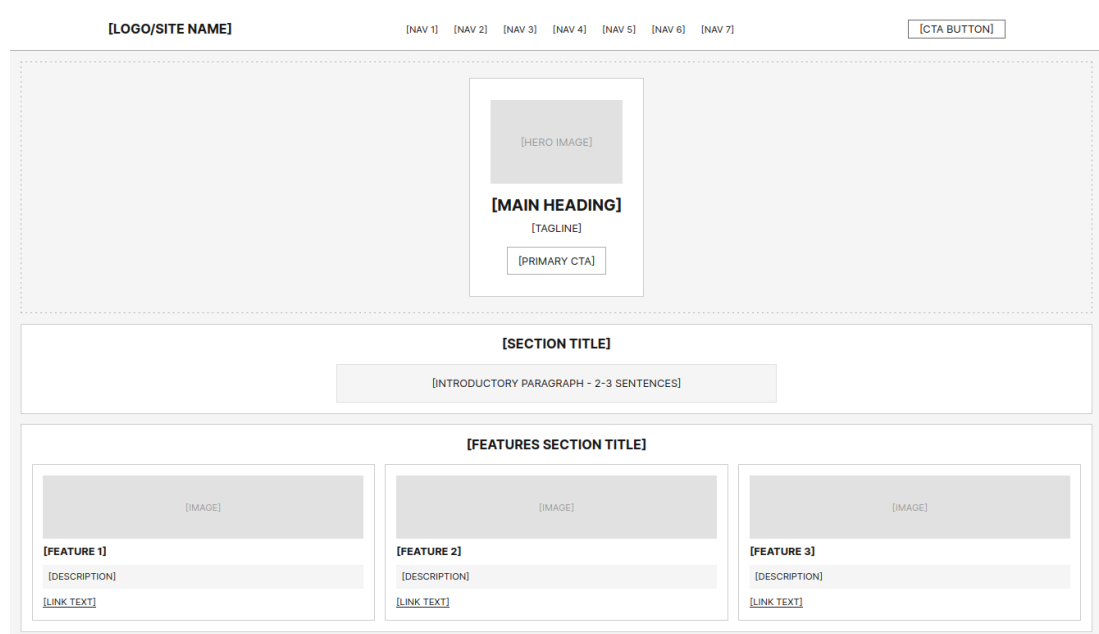


Figure 4.6: Low-Fidelity Wireframe: Homepage

4.4.2 Photo Gallery

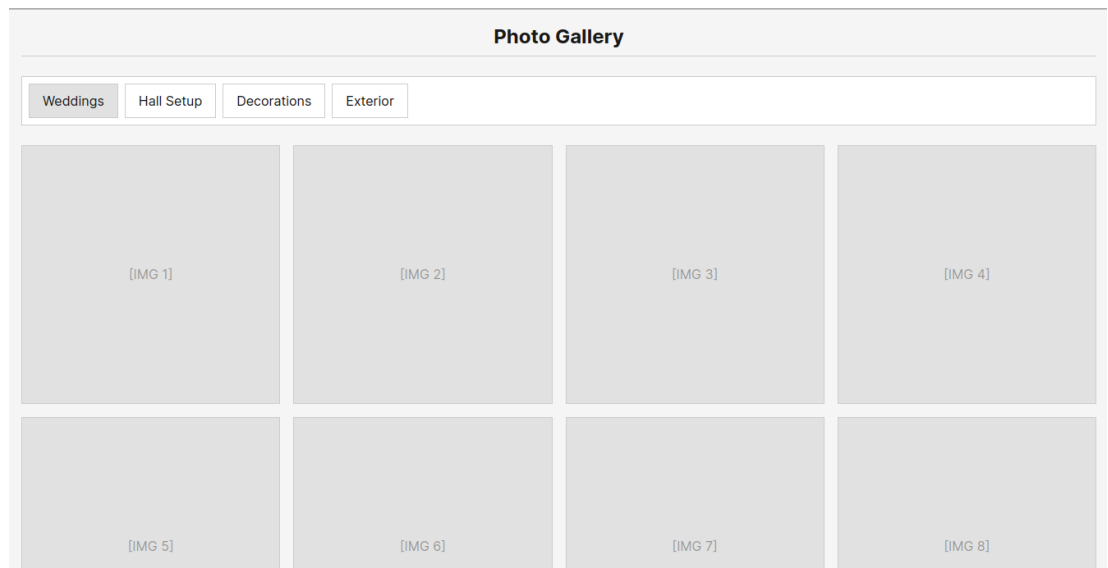


Figure 4.7: Low-Fidelity Wireframe: Gallery Page

4.4.3 Availability calendar

Check Availability

<
May 2025
>

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

☐ Available
 ☐ Pending
 ☐ Booked

Make an Inquiry

Figure 4.8: Low-Fidelity Wireframe: Availability Calendar Page

4.4.4 Pricing Page

[PAGE TITLE]

[INTRODUCTORY TEXT BLOCK]

[PACKAGE 1]

[PRICE]

- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]

[BUTTON TEXT]

[PACKAGE 2]

[PRICE]

- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]

[BUTTON TEXT]

[PACKAGE 3]

[PRICE]

- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]
- ✓ [FEATURE]

[BUTTON TEXT]

[SECTION TITLE]

[SERVICE NAME]

[PRICE INFO]

[SERVICE NAME]

[PRICE INFO]

[SERVICE NAME]

[PRICE INFO]

[SERVICE NAME]

[PRICE INFO]

[SERVICE NAME]

[PRICE INFO]

[SERVICE NAME]

[PRICE INFO]

[CTA BUTTON]

Figure 4.9: Low-Fidelity Wireframe: Pricing Page

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4.4.5 Inquiry Form Page

[PAGE TITLE]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
mm / dd / yyyy

[FIELD LABEL]
[SELECT OPTION]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[SELECT OPTION]

[PRIVACY/DISCLAIMER TEXT]

[SUBMIT BUTTON]

[SIDEBAR TITLE]

[CONTACT METHOD]: [CONTACT INFO]

[CONTACT METHOD]: [CONTACT INFO]

[INFO TITLE]:
[INFO LINE 1]
[INFO LINE 2]

Figure 4.10: Low-Fidelity Wireframe: Inquiry Form Page